

Culturally responsive AI chatbots: From framework to field evidence

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ABSTRACT

As AI systems become part of everyday life around the world, their failure to recognise and respond to cultural differences can erode trust, reduce engagement, and undermine legitimacy. This paper introduces the Culturally Responsive Artificial Intelligence (Chatbot) Framework (CRAIF-C), a practical, modular approach to building AI chatbots that understand and respect cultural diversity. CRAIF-C is novel in that it operationalises cultural responsiveness across the *entire* AI lifecycle, combining domain-specific technical methods with validated measurement tools and multi-context empirical testing. It addresses persistent limitations of earlier approaches, such as Value-Sensitive Design or Participatory AI, which often remain conceptual, sector-bound, or late-stage interventions. CRAIF-C works across four key domains: Enculturation, Adaptive Interaction, Explainability & Transparency, and Governance & Accountability. The framework's effectiveness is demonstrated through four complementary studies, which consistently show that AI chatbot systems using CRAIF-C achieve meaningful gains in cultural fit, natural communication, clear explanations, user trust, and sustained engagement. By incorporating cultural sensitivity into the core of AI chatbot design, CRAIF-C provides a roadmap for creating technology that is both technically capable, socially aware, ethically robust, and globally adaptable.

1. Introduction

Businesses are increasingly deploying artificial intelligence (AI)-powered conversational agents, such as chatbots, across multicultural markets as part of their customer engagement strategies, since conversational AI provides a cost-effective way to deliver 24/7 service, manage high-volume inquiries, and personalise interactions at scale (Roy & Naidoo, 2021). However, as these systems transition from single-market pilots to geographically diverse rollouts, they encounter users with heterogeneous cultural expectations, language norms, and communication styles. What counts as polite, persuasive, or trustworthy in one market may be interpreted as abrupt, ineffective, or even offensive in another. As a result, the imperative for AI chatbot designs that are inclusive, context-aware, and culturally responsive has become increasingly urgent (Barnes et al., 2024). In response, algorithmic bias scholarship has rapidly expanded in recent years (Barocas et al., 2019; Mehrabi et al., 2021; Prabhakaran et al., 2022), focusing predominantly on disparities related to race, gender, and socioeconomic status. However, cultural diversity, encompassing shared values, beliefs, norms, languages, symbolic systems, and communication practices, remains a persistent and underexamined blind spot in AI research and practice

(Costanza-Chock, 2020; De Freitas et al., 2023; Ge et al., 2024). The neglect of cultural variables in AI design is not a peripheral oversight; it fundamentally undermines system performance, user trust, and long-term adoption (Blodgett et al., 2020; Crawford, 2021). AI systems that fail to account for cultural norms risk producing outputs that feel alien or misaligned, leading to exclusion, misrepresentation, reduced legitimacy, and in some cases active harm (Benjamin, 2019; Birhane, 2021; Eubanks, 2018).

This problem is well-documented in the field of Human-Computer Interaction (HCI), where past work has underscored the role of culture in shaping not only user satisfaction but also cognitive load, trust calibration, and decision-making efficacy (Clemmensen et al., 2007; Marcus & Gould, 2000; Yuan et al., 2023). Similarly, the cross-cultural literature through theoretical models such as Hofstede's cultural dimensions (Hofstede, 2001), Triandis' individualism-collectivism framework (Triandis, 1995), and Hall's high- and low-context communication theory (1976) offer foundational insights into cross-cultural variation in cognition, behaviour, and communication style. Moreover, participatory design (e.g. Irani & Silberman, 2016), value-sensitive design (e.g., Friedman et al., 2013), and inclusive design principles (e.g. Clarkson et al., 2003) have long advocated embedding the lived experiences and

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values of diverse communities into technology development. Yet, despite these advances, the integration of cultural responsiveness into AI systems remains fragmented, inconsistent, and often tokenistic (Barnes et al., 2024; Mehrabi et al., 2021; Prabhakaran et al., 2022). Indeed, most AI pipelines, from dataset curation and annotation to model training, deployment, and explainability, operate within monocultural paradigms implicitly aligned with Western epistemologies and cognitive schemas (Birhane, 2021; Blodgett et al., 2020). As AI continues to penetrate markets and governance structures in regions with vastly different cultural traditions, these narrow defaults create predictable mismatches between system behaviour and user expectations (Jobin et al., 2019; Crawford, 2021). Such mismatches not only limit user engagement but can also exacerbate digital inequities by privileging certain cultural logics over others.

Emerging studies are beginning to expose the limitations of culturally naïve AI. For example, Singh et al. (2025) proposed an approach to modelling culturally adaptive AI agents, while simultaneously noting the absence of scalable, real-world deployments. Peters and Carman (2024) demonstrated that widely used explainable AI frameworks are grounded in Western analytic reasoning styles, potentially alienating users from holistic or relational cultures. Parag et al. (2023) introduced a framework for culturally inclusive AI in healthcare, but its application remains restricted to small-scale, sector-specific trials. Other work in cross-cultural HCI (e.g., Reinecke & Bernstein, 2013) and human-centred AI (e.g., Yuan et al., 2023) reinforces that cultural responsiveness is not a luxury. It is a precondition for ethical, useable, and sustainable AI.

This paper responds to that gap by introducing the Culturally Responsive Artificial Intelligence (Chatbot) Framework (CRAIF-C). CRAIF-C is novel in that it integrates cultural awareness at *every stage* of AI chatbot design and deployment, from data curation and model architecture to explanation strategies and governance mechanisms, through domain-specific, technically implementable methods. It moves beyond aspirational principles by embedding measurable, scalable practices tested across diverse cultural contexts and sectors. In doing so, CRAIF-C reframes cultural responsiveness as a primary design principle and a measurable driver of trust, engagement, adoption, and long-term system legitimacy.

Against this backdrop, our primary objective in this paper is to confront the persistent gap between the cross-cultural realities of contemporary AI deployment and predominantly language-centric design and evaluation practices; a gap that yields predictable failure modes, including mis-calibrated tone and politeness, culturally insensitive outputs, context-dependent fairness harms, and explanation styles misaligned with local reasoning norms (Ryström et al., 2025). We define and operationalise CRAIF-C as a full lifecycle framework that bridges the gap between multilingual capability and multicultural alignment by specifying concrete practices spanning requirements, data curation, model development, interaction design, evaluation, and post-deployment monitoring. Our secondary objective is to demonstrate measurable improvements on culture-relevant outcomes (usability, trust, and error profiles) through preregistered tests and ablations that isolate CRAIF-C's contribution. In addressing long-standing challenges (e.g., translation versus true localisation, context-dependent fairness, explanation-style mismatch, and cultural drift after launch), CRAIF-C treats cultural responsiveness not as an optional enhancement but as a set of auditable controls with clear decision rules and escalation paths. The remainder of the paper follows this logic: we motivate why culture must be a first-class design requirement, formalise CRAIF-C and its artefacts, empirically test their impact and robustness, and translate those results into deployment guidance. In summary, our objectives and contributions converge on a single promise to practitioners and scholars alike: how to maintain AI chatbots' effectiveness in culturally diverse, real-world settings.

2. Literature review

A substantial body of psychological and cross-cultural research has shown how culture shapes human cognition, perception, and decision-making (Chiu & Hong, 2006; Nisbett et al., 2001). Yet despite the urgency of these insights, mainstream AI development continues to privilege a culturally narrow worldview (Peters & Carman, 2024). Indeed, a growing body of interdisciplinary research has critically examined the cultural biases embedded in AI systems, particularly within natural language processing (NLP), recommender systems, and human-computer interaction (Reinecke & Bernstein, 2013; Bansal, 2022; Gupta et al., 2022). Scholars have consistently pointed to the ways in which AI encodes dominant cultural norms, often as a result of imbalanced training datasets, Anglocentric user experience (UX) standards, and insufficient localisation during the development pipeline. These concerns are not only theoretical. Empirical work by Luna and Forquer Gupta (2001) demonstrates that culturally incongruent communication diminishes user engagement and trust, findings echoed across studies examining cross-cultural digital interaction (Reinecke & Bernstein, 2013).

Several frameworks have sought to address cultural inclusivity in AI, but most remain at the level of broad ethical aspirations rather than technically grounded design principles. For example, Hall's (1976) theory of high-versus low-context communication continues to underpin many UX localisation practices. However, such ideas have rarely been operationalised across full AI deployment cycles. Similarly, more recent contributions by Serrano and Smith (2019) and Wolf et al. (2020) have turned to explainability and fairness, but cultural adaptability as a core design criterion remains largely overlooked. Even as fairness-aware algorithms proliferate, interventions aimed at cultural sensitivity are typically piecemeal and post hoc, such as the addition of multilingual options or region-specific heuristics, rather than being embedded from the outset.

Participatory AI (Irani & Silberman, 2016) has also emerged as a corrective intervention, involving communities in system design. However, participation often occurs late in the development cycle, when key architectural choices are already locked in, limiting its impact on core technical features. For example, Qin et al. (2023) discuss how this timing issue is especially problematic in global health and development contexts, where AI systems developed in high-income countries are deployed in low- and middle-income countries with minimal adaptation.

The importance of aligning technology with cultural values, cognitive styles, and interaction expectations is well established in HCI. For instance, Marcus and Gould's (2000) cultural usability frameworks and Clemmensen et al.'s (2007) culturally sensitive heuristics demonstrate that culture shapes usability and adoption. Yet, the underlying logic of most AI models still reflects Western, individualist reasoning styles. For example, large datasets such as ImageNet are skewed toward Western object categories, and language models like BERT and GPT are trained predominantly on English-language corpora. This bias extends to interpretive layers with generative AI often reproducing cultural stereotypes in depictions of professionalism, beauty, intelligence, and social roles (Bender et al., 2021).

Fairness research in AI is another stream of the literature that has sought to address inclusivity, but fairness definitions often reflect universalist assumptions. Distributive fairness, for example, is emphasised in individualist societies, while procedural or relational fairness may hold greater importance in collectivist or high-context cultures (Binns et al., 2018; Fjeld et al., 2020; Kim & Markus, 1999; Whittaker et al., 2018). Similarly, common explainability tools such as LIME and SHAP rely on feature-attribution models grounded in analytic reasoning styles, which may conflict with preferences for narrative or holistic explanations in high-context cultures (Miller, 2019).

Furthermore, efforts like Participatory AI (e.g., Magaña & Shilton, 2025) have made progress, but even these can be constrained by the practicalities of scaling culturally embedded design. CRAIF-C

distinguishes itself by combining early-stage cultural integration with technical specificity and validated measurement instruments, delivering a lifecycle-wide approach rather than one-off adjustments. Table 1 compares CRAIF-C with widely cited frameworks, acknowledging their substantive contributions while clarifying where CRAIF-C extends their scope.

In sum, while the literature provides valuable conceptual, ethical, and methodological starting points, there is no comprehensive framework that integrates cultural responsiveness as a technically grounded, measurable, and scalable design logic throughout the AI lifecycle. CRAIF-C is suggested to fill this gap, and is organised around four interconnected domains: Enculturation, Adaptive Interaction, Explainability & Transparency, and Governance & Accountability which are outlined next.

3. From framework to function: operationalising CRAIF-C

As outlined in the previous section, existing scholarship has identified multiple shortcomings in how AI frameworks address cultural diversity, ranging from high-level ethical principles that lack technical specificity (Costanza-Chock, 2020; Friedman et al., 2013) to explainability models that encode monocultural reasoning styles (Peters & Carman, 2024; Yuan et al., 2023) and governance mechanisms that exclude community perspectives (Birhane, 2021; Jobin et al., 2019). Building on these observations, the Culturally Responsive Artificial Intelligence (Chatbot) Framework (CRAIF-C) is proposed to address five recurrent and interlinked weaknesses in current AI design and governance:

1. *The formalisation gap*: While cultural awareness is often acknowledged in principle, it is rarely operationalised into precise requirements that can be embedded in data curation, model development, or interface design. This echoes the critique that cultural considerations remain aspirational rather than actionable in most AI ethics guidelines (Jobin et al., 2019).
2. *The interpretability gap*: Explanation models frequently privilege Western analytic reasoning, creating a mismatch with the interpretive norms of users from other cultural backgrounds (Peters & Carman, 2024; Yuan et al., 2023).
3. *The governance gap*: Oversight structures seldom include culturally informed decision-making processes, perpetuating the exclusion of

affected communities from AI governance (Birhane, 2021; Costanza-Chock, 2020).

4. *The evaluation gap*: Performance metrics and benchmarks typically ignore cultural alignment, focusing instead on universalised accuracy or efficiency measures that fail to capture user trust or contextual fit (Blodgett et al., 2020).
5. *The integration gap*: When cultural factors are considered, they are often introduced late in development as isolated fixes, making them difficult to integrate systematically or scale sustainably, an issue noted across participatory design and inclusive design scholarship (Friedman et al., 2013; Persson et al., 2015).

While these gaps appear distinct, they share a common root cause: cultural responsiveness throughout the AI lifecycle, spanning data, model, interaction, and governance. CRAIF-C is proposed to go beyond statements of intent or one-off inclusivity features. It works as a practical methodology, giving teams clear steps they can follow, test, and adapt. Grounded in design science research (DSR) principles, CRAIF-C turns high-level goals into technical methods that can be replicated across projects. The aim is straightforward: to create AI systems that are culturally attuned from the start and stay that way throughout their use. The framework's novelty lies in its combination of:

- Full lifecycle integration: from data curation to governance.
- Technical specificity: operational methods for each domain, not just conceptual guidance.
- Validated measurement instruments: ensuring cultural responsiveness can be assessed consistently.
- Empirical generalisability: tested in small start-ups, multi-country experiments, and year-long deployments.

The CRAIF-C is organised around four interconnected domains: Enculturation, Adaptive Interaction, Explainability & Transparency and Governance & Accountability. Together, these domains make cultural responsiveness a foundational design element rather than a late-stage modification.

The first domain of the CRAIF-C, Enculturation, focuses on closing the formalisation gap by ensuring that cultural variables are built into training data, model logic, and user experience design. Techniques such as multilingual validation, contextual annotation, and curating cultural narratives strengthen this foundation. For example, back-translation using models like MarianMT, XLM-R, and mBART can reveal when

Table 1
Comparative positioning of CRAIF-C and existing design approaches.

Framework	Primary Focus/ Scope	Cultural Integration	Technical Specificity	Lifecycle Coverage	Measurement Tools	Scalability	Empirical Validation	Limitations
Value-Sensitive Design (e.g., Friedman et al., 2013)	Embeds stakeholder values into technology design	Indirect; culture addressed through broad values	Conceptual guidance, minimal technical methods	Primarily early-stage design and evaluation	Limited, context-dependent	Conceptually scalable but rarely tested in global deployments	Select case studies	Limited operational detail for cultural factors
Participatory AI (e.g., Magaña & Shilton, 2025)	Involves affected communities in AI development	Culture addressed via community input	Process-focused, not technical	Often concentrated in early design and testing phases	Varies widely, no standardised tools	Scaling depends on ongoing community involvement	Select case studies	Often applied late; lacks integration into model architecture
Fairness-Aware ML (e.g., Barocas et al., 2019)	Reduces algorithmic bias	Cultural differences rarely included	High technical detail for statistical bias mitigation	Model training and evaluation	Bias metrics (e.g., demographic parity)	Scalable for statistical bias checks	Established in bias literature	Narrow fairness metrics, often universalist
Culturally Responsive AI Framework across the full AI lifecycle (CRAIF-C) (this paper)	Embeds cultural responsiveness	Direct integration into data, interaction, explainability, and governance	Domain-specific, technically implementable methods in four areas	From data collection and model development to deployment, governance, and adaptation	Validated scales and coding rubrics for cultural responsiveness	Modular architecture tested in cross-cultural, multi-country contexts	Four interlinked empirical studies across ethnographic, experimental, and longitudinal designs	Requires interdisciplinary teams; adoption curve for new methods

meaning changes in translation. A phrase like “speak your mind” might sound empowering in a low-context culture but could come across as impolite in a collectivist one. Detecting and fixing these mismatches early helps the system stay culturally accurate. In computer vision, working with culturally tagged corpora and local experts ensures that symbols, clothing, or rituals are understood as intended.

The second domain, Adaptive Interaction, addresses the evaluation and integration gaps by teaching AI systems to respond to cultural cues in real time. For example, reinforcement learning models can be adapted to include culturally informed reward functions, allowing systems to adjust tone, pacing, or politeness based on the user’s cultural profile. Storytelling bots can also be fine-tuned to follow circular narrative structures common in communal cultures or Western linear plots, depending on the audience. In multimodal systems, methods such as neural style transfer and metadata tagging make it possible to tailor images, text, and audio so they feel culturally relevant.

The third domain, Explainability and Transparency, responds to the interpretability gap by offering explanations that fit different ways of thinking. Many existing explainability tools, like LIME (Local Interpretable Model-agnostic Explanations, e.g., Ribeiro et al., 2016) or SHAP (SHapley Additive exPlanations, e.g., Lundberg & Lee, 2017), rely on Western-style analytical logic. CRAIF-C encourages the option to choose explanation formats (e.g., statistical, logical, or narrative) so users can pick what makes the most sense to them. For high-context cultures, a story or metaphor might be more effective than a step-by-step breakdown. Similarly, in visual systems, Class Activation Maps can be labelled in culturally meaningful terms, while NLP explanations can be linked to familiar idioms.

The fourth domain, Governance and Accountability, tackles the governance gap by making cultural perspectives part of decision-making. For instance, Cultural Impact Assessments (CIAs) can be carried out before deployment to evaluate cultural risks, with these processes co-led by cultural co-design panels. Moreover, model cards and datasheets (Geburu et al., 2018; Mitchell et al., 2019) can be extended to include metrics on cultural representation, diversity of data sources, and inclusivity of review panels. Governance structures should mix technical expertise with deep cultural knowledge.

Each of these domains maps to a specific gap identified in the literature (see Table 2) and is backed by methods that can be integrated into everyday workflows. When applied together, they offer a systematic way to build AI systems that are both technically robust and culturally responsive. When viewed alongside existing approaches, CRAIF-C stands out for its combination of cultural scope, technical detail, and lifecycle-wide application. Appendix 1 translates CRAIF-C into a one-page, sprint-ready checklist, mapping each domain to concrete artefacts, steps, KPIs, risks/controls, and governance docs, so teams can implement, audit, and iterate cultural responsiveness across the

lifecycle. The studies that follow show how CRAIF-C has been implemented in different contexts, using ethnographic, experimental, and longitudinal approaches to test its feasibility, effectiveness, and long-term potential.

4. Methodology: testing CRAIF-C

To evaluate CRAIF-C both as a theoretical framework and as a practical toolkit, this research employed a multi-method, multi-context design. The aim was to explore how cultural responsiveness in AI Chatbots can be implemented, experienced, and measured across multiple stakeholder levels. Four studies were conducted, each offering a distinct perspective on the framework’s application.

4.1. Study 1: a pre- and post-implementation in AI development environment

Study 1 examines how CRAIF-C was applied in a lean AI development environment that had no formal frameworks for cultural adaptation. The setting was a small Sydney (Australia)-based start-up, referred to here as PolyglotAI (a pseudonym for confidentiality reason). PolyglotAI builds multilingual chatbots primarily for Southeast Asian markets. While the company’s systems were technically sound, they were culturally brittle. End-user engagement with their AI systems was steadily declining, and users consistently reported problems with tone, idiomatic phrasing, and trust. Part of the company’s challenge was that PolyglotAI’s small team of internal developers considered cultural fit a linguistic challenge rather than a core design requirement.

To address these challenges, PolyglotAI invited one of the authors to provide targeted CRAIF-C training. This was not a one-off workshop, but a sustained, embedded intervention designed to help the team integrate cultural responsiveness from the start of development. Over six months, PolyglotAI’s small team consisting of two machine learning engineers, two software developers, one UX designer, and one data ethicist took part in a combination of intensive workshops and hands-on mentoring. These sessions were timed to coincide with live development sprints so that new approaches could be tested in real time.

The aim was to make CRAIF-C part of the team’s daily workflow, fostering a shift in mindset, governance, and collaborative practice. Data collection involved more than 120 h of ethnographic observation during design sprints, 14 semi-structured interviews, and the analysis of 48 artefacts such as sprint logs, wireframes, and code documentation.

The coding process of this collected data followed a grounded theory approach across three stages. In the first stage, open coding was applied line by line to all transcripts, field notes, and artefacts, generating 87 first-order codes that captured recurring patterns in language, behaviours, and decision-making. Two researchers coded the data

Table 2
Addressing key gaps in culturally responsive AI through CRAIF-C.

Identified Gap in Literature	Description	Relevant CRAIF-C Domain(s)	How CRAIF-C Addresses the Identified Gaps?
1. Formalisation Gap	Lack of computational models and design logics that encode cultural variables into AI architecture	Enculturation; Adaptive Interaction	Provides guidelines for embedding cultural schemas, values, and semiotics during data curation, model training, and interface development
2. Interpretability Gap	Explainable AI (XAI) tools are rooted in Western analytic reasoning, misaligned with users from high-context or holistic reasoning cultures	Explainability and Transparency	Recommends culturally attuned explanation strategies (e.g., narrative, metaphorical, or relational forms of transparency)
3. Governance Gap	Cultural worldviews are underrepresented in AI governance structures and design authority	Governance and Accountability	Promotes inclusive, pluralistic governance models, including co-design, community-led audits, and culturally diverse ethics panels
4. Evaluation Gap	Absence of validated metrics to assess cultural alignment or responsiveness	All domains (especially Adaptive Interaction and Explainability)	Introduces evaluative criteria and performance benchmarks that reflect cultural relevance, user trust, and contextual alignment
5. Integration Gap	Cultural considerations remain siloed or reactive, often confined to specific sectors or post hoc adjustments	Entire CRAIF-C Framework	Offers a cross-sectoral, lifecycle-based roadmap to integrate culture from conception to deployment across domains, applications, and institutions

independently in NVivo and then met regularly to reconcile differences, reaching agreement before moving forward. In the second stage, axial coding grouped related codes into 12 thematic clusters (see [Appendix 2](#)), focusing on how technical decisions intersected with cultural considerations. In the final stage, selective coding distilled these clusters into six core themes: Shifting Design Consciousness, Cultural Empathy in Code, Affective Transformation, Modularisation for Sustainability, Localisation versus Scalability, and Governance Through Partnerships (see [Table 3](#)). These themes were mapped to CRAIF-C's four domains, with Governance Through Partnerships directly addressing the lack of oversight structures at the start of the project. Cohen's Kappa at the axial coding stage was 0.84, indicating strong inter-rater reliability.

The intervention led to concrete technical changes. For example, the team introduced multilingual back-translation protocols using MarianMT for Bahasa Indonesia, Vietnamese, and Thai. They enriched the NLP pipeline with culturally specific corpora that included regional folklore and consumer product reviews, and they implemented culture-aware attention masks so that transformer-based models could prioritise culturally relevant language patterns. This change was reflected in the sprint logs. From 1243 chatbot utterances, 214 (17.2 percent) were flagged for cultural incongruence. These often involved semantic drift, overly direct phrasing, or idioms that lacked meaning in the target culture. Revisions were made with input from regional consultants, and updated material was reintegrated into the training corpus.

The six PolyglotAI staff members were also invited to complete a survey at both the start and conclusion of the intervention, capturing their perceptions of how the four CRAIF-C domains would improve end-user experiences. As discussed previously, each domain reflects a distinct yet interconnected dimension of user experience. Enculturation represents perceived cultural fit because it captures the extent to which the AI system aligns with the user's cultural norms, values, and linguistic preferences. When these elements are present, users feel recognised and understood, fostering a sense of belonging. Adaptive Interaction reflects communication naturalness, as it describes the AI's ability to adjust its tone, pacing, and content dynamically in response to user cues, producing a conversational flow that feels fluid and human-like. Explainability and Transparency can be expressed as explanation resonance; the degree to which the AI's explanations not only clarify its reasoning but also connect meaningfully with the user's prior knowledge, context, and decision-making needs. Finally, Governance and Accountability can translate into trust in recommendations, as these mechanisms signal that outputs are generated ethically, responsibly, and with safeguards against harm, thereby reinforcing user confidence.

Enculturation, Adaptive Interaction, Explainability and Transparency, and Governance and Accountability were each measured using five items adapted from validated scales for chatbot contexts (see [Appendix 3](#); $\alpha = .82\text{--}0.88$). These four domains collectively set the stage for Affective Engagement, or user engagement, which emerges when interactions are culturally aligned, communicatively natural, meaningfully explained, and demonstrably trustworthy, making it the emotional and behavioural manifestation of these underlying design principles working in concert. Items were rated on a five-point Likert scale from strongly disagree to strongly agree.

The results showed statistically significant improvement in every domain. Enculturation increased from a mean of 3.40–4.32 ($t(5) = 5.16$, $p < .01$). Adaptive Interaction rose from 3.29 to 4.38 ($t(5) = 4.75$, $p < .01$). Explainability and Transparency improved from 3.25 to 4.45 ($t(5) = 4.87$, $p < .01$). Governance and Accountability increased from 3.39 to 4.28 ($t(5) = 4.62$, $p < .01$). These results demonstrate that CRAIF-C functions as both a conceptual model and a practical toolkit. When cultural responsibility is shared across a team and reinforced through governance structures and continuous reflection, inclusivity becomes an everyday practice rather than an afterthought.

Furthermore, network analysis of sprint documentation and version control logs supported these findings and showed a rise in cross-role engagement. At the beginning of the project, cultural considerations

were mostly confined to the UX role, with the UX designer acting as the main knowledge broker (degree centrality = 0.72). Adaptive Interaction factors were rarely discussed in technical meetings, Explainability had no shared criteria, and Governance was absent. By the third month of implementation, network density had increased from 0.31 to 0.58, showing more cross-functional engagement. Closeness centrality became more evenly distributed, with the highest score dropping from 0.81 to 0.67 and the lowest rising from 0.42 to 0.59. Governance and Accountability was now embedded across all roles, supported by fairness metrics and review checkpoints co-designed with external stakeholders (see [Fig. 1](#)).

Overall, the PolyglotAI case study shows that the implementation of CRAIF-C can deliver measurable improvements in cultural responsiveness, explanation resonance, and trust. Qualitative themes such as Cultural Empathy in Code and Affective Transformation pointed to deeper shifts in how developers approached their work, reframing error diagnosis through a cultural lens and using inclusive metaphors in design discussions. Technical measures such as back-translation, enriched corpora, and culture-aware attention masks proved compatible with modern NLP architectures without creating prohibitive costs. High inter-rater reliability in coding confirmed that assessments of cultural congruence were consistent and reproducible. The findings from Study 1 are summarised in [Table 4](#).

4.2. Study 2: controlled user testing of CRAIF-C enhancements

While Study 1 examined how CRAIF-C was embedded internally within a development team, Study 2 shifted focus to actual end users. The aim was to test whether the four CRAIF-C domains (Enculturation, Adaptive Interaction, Explainability and Transparency, and Governance and Accountability) could produce measurable improvements in user experience during direct chatbot interaction.

A total of 354 participants were recruited from three target cultural markets: Indonesia ($n = 120$), Vietnam ($n = 117$), and Thailand ($n = 117$). Participants were drawn from an ISO-compliant online panel with device/geo checks and duplicate detection. Eligibility required age ≥ 18 , fluency in the target locale, and ≥ 1 AI/chatbot interaction in the past six months). Recruitment of participants also applied quotas to balance gender, age, and prior chatbot experience. We determined the minimum required sample size via an a priori power analysis (two-tailed $\alpha = .05$, 80 % power) to detect a small-to-moderate standardised effect ($d \approx 0.30$), inflating targets by $\sim 10\text{--}15\%$ to accommodate exclusions; our achieved sample ($N = 354$) met these criteria. The participants were randomly assigned to interact with either a baseline chatbot, which had no cultural enhancements, or a CRAIF-C enhanced chatbot. These enhancements included culturally tailored opening sequences, narrative-based explanations, adaptive conversational pacing, and explicit trust cues. All interactions occurred in a secure apparel e-commerce shopping sandbox that mirrored production customer-journey workflows (e.g., product discovery and comparison, size/fit guidance, promotions and checkout, order status and returns).

After a guided shopping simulation involving the chatbot, participants completed a post-interaction survey. The survey measured the four CRAIF-C domains plus affective engagement, using the same adapted validated scales used in Study 1 (see [Appendix 3](#)). Reliability was strong (Cronbach's $\alpha \geq .8$), and confirmatory factor analysis confirmed validity (CFI = 0.95, RMSEA = 0.042, AVE > 0.70).

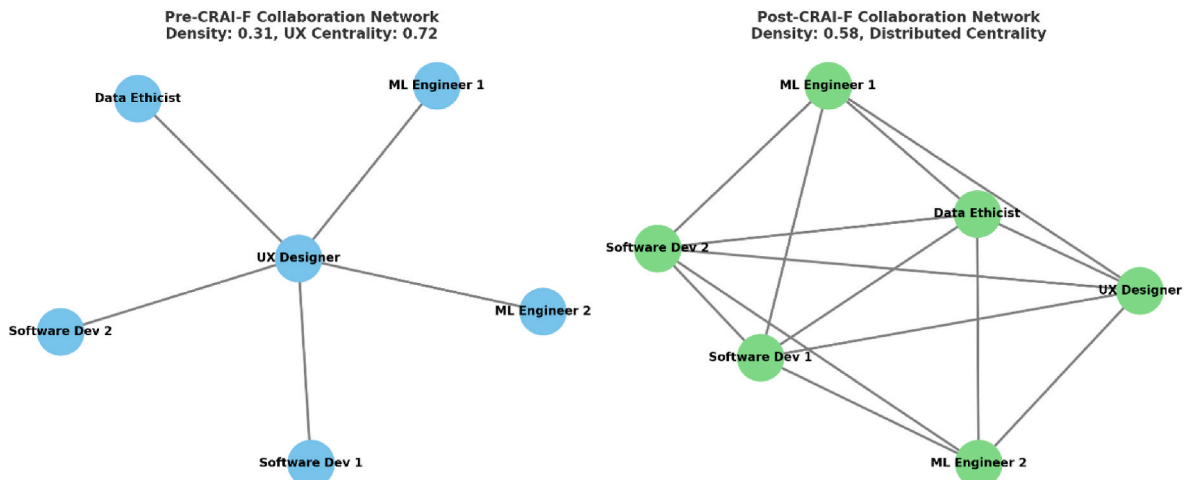
We next conducted a MANCOVA¹ test controlling for age, gender, and prior chatbot experience. The observed results showed a statistically

¹ The MANCOVA tests whether the CRAIF-C domains, considered jointly, shift the pattern of usability/trust/pragmatics outcomes relative to Baseline. Effect sizes (partial η^2) quantify how much improvement each domain yields; Confidence Intervals express precision; P-values complement interpretation by indicating how inconsistent the data are with no true difference.

Table 3

Thematic Consolidation from axial themes to core constructs.

Final Core Theme	Merged Axial Themes	Data Co-occurrence Rate	# of Supporting First-order Codes	Associated CRAIF-C Domain(s)
Shifting Design Consciousness	Design Ownership Diffusion; Code-Level Resistance and Buy-in	71 % of sprint debriefs	14	Enculturation
Cultural Empathy in Code	Cultural Debugging; Reframing Error Diagnosis; Inclusive Metaphor Use	74 % of reflective interviews	13	Governance & Accountability
Affective Transformation	Emotion-Driven Design; Reflexive Design Moments; Affective Accountability	65 % of emotional narratives	12	Enculturation
Modularisation for Sustainability	Patterned Prompt Design; Scalable Template Logic	68 % of code audit entries	11	Adaptive Interaction
Localisation vs. Scalability	UX Localisation Constraints; Resource Allocation Strategies	70 % of sprint reflections	10	Adaptive Interaction
Governance Through Partnerships	Community Vetting; Stakeholder Consultation Feedback Loops	60 % of workshop transcripts	9	Governance & Accountability

**Fig. 1.** Social network analysis (pre and post CRAIF-C deployment).**Table 4**

Summary of CRAIF-C implementation outcomes in PolyglotAI (study 1).

CRAIF-C Domain	Key Changes Implemented	Quantitative Outcomes	95 % CI for Mean Difference	Qualitative Themes	Structural/Process Changes
Enculturation	Multilingual back-translation protocols (MarianMT for Bahasa Indonesia, Vietnamese, and Thai); culturally enriched corpora with regional folklore and product reviews; culture-aware attention masks in NLP models	Mean increased from 3.40 to 4.32, $t(5) = 5.16$, $p < .01$	[0.57, 1.27]	Cultural Empathy in Code; Shifting Design Consciousness	Contextual annotation integrated into sprint cycles; local expert consultation
Adaptive Interaction	Real-time tone and pacing adjustments; culturally specific narrative structures in dialogue flows	Mean increased from 3.29 to 4.38, $t(5) = 4.75$, $p < .01$	[0.51, 1.67]	Localisation versus Scalability; Affective Transformation	Reinforcement learning reward functions adapted for cultural responsiveness
Explainability and Transparency	Narrative-based and metaphor-rich explanations; culturally meaningful labelling of Class Activation Maps	Mean increased from 3.25 to 4.45, $t(5) = 4.87$, $p < .01$	[0.58, 1.82]	Shifting Design Consciousness; Affective Transformation	Co-created explanation styles with regional stakeholders
Governance and Accountability	Cultural Impact Assessments; fairness metrics; cultural review checkpoints	Mean increased from 3.39 to 4.28, $t(5) = 4.62$, $p < .01$	[0.46, 1.32]	Governance Through Partnerships	Governance embedded across all roles; decentralised oversight through stakeholder panels

significant overall model, Wilks' Lambda = 0.68, $F(5, 346) = 31.27$, $p < .001$, $\eta^2 = 0.31$. Follow-up univariate tests revealed consistent advantages for the CRAIF-C chatbot across all outcomes. Adaptive Interaction scores were higher for the CRAIF-C chatbot ($M = 4.28$, $SD = 0.64$) compared with the baseline ($M = 3.61$, $SD = 0.71$), a mean difference of 0.67 ($F = 45.13$, $p < .001$, $\eta^2 = 0.12$). Governance and Accountability also improved ($M = 4.11$ vs. 3.45), $F = 38.92$, $p < .001$, $\eta^2 = 0.10$. The largest effects were for Explainability and Transparency (mean difference 1.16, $F = 62.84$, $p < .001$, $\eta^2 = 0.15$), and Enculturation (mean

difference 1.15, $F = 59.77$, $p < .001$, $\eta^2 = 0.14$). Affective Engagement also showed a meaningful gain (mean difference 0.84, $F = 40.16$, $p < .001$, $\eta^2 = 0.11$). To test generalisability, we computed subgroup contrasts by locale (Indonesia, Vietnam, Thailand), age (18–29, 30–44, 45+), and gender, reporting Hedges' g with 95 % CIs and exact two-sided p -values ($\alpha = .05$). Arm $\times$ Locale, Arm $\times$ Age, and Arm $\times$ Gender interaction tests in follow-up ANCOVAs (HC3) showed no significant heterogeneity after Holm adjustment (all $p_{adj} > 0.05$), with subgroup CIs overlapping the overall effects. Safety (a standard measure

in AI/HCI literature that allows an investigation into whether CRAIF-C improves cultural fit without making the system less truthful or more harmful) was held within each subgroup, factuality non-inferiority was satisfied ($\Delta \leq 0.02$, one-sided 95 % CIs), and no fairness degradation was detected.

Furthermore, mediation analysis indicated that explanation resonance mediated the link between chatbot condition and trust ($b = 0.36$, 95 % CI [0.25, 0.48]), supporting CRAIF-C's proposition that culturally aligned explanations build trust by increasing epistemic comfort. Moderation analysis also showed that participants with higher cultural familiarity engaged more deeply with the CRAIF-C chatbot, $F = 27.6$, $p < .001$.

These results confirm that CRAIF-C's domains function as intended at the user level. Enculturation improved perceived fit, Adaptive Interaction enhanced conversational naturalness, Explainability and Transparency boosted explanation resonance, and Governance and Accountability improved trust. Gains in affective engagement suggest that when these domains operate together, they not only improve usability but also deepen emotional connection with the system.

4.3. Study 3: experimental evaluation of CRAIF-C components

Following the controlled end-user evaluation in Study 2, Study 3 aimed to test each CRAIF-C domain individually to determine its unique and combined impact on end-user experience. The four domains (Enculturation, Adaptive Interaction, Explainability and Transparency, and Governance and Accountability) were examined against a baseline chatbot. Each was measured using the same validated scales used in the previous studies (see Appendix 3). A total of 750 participants were recruited evenly from Australia ($n = 150$), India ($n = 150$), the Philippines ($n = 150$), and Vietnam ($n = 150$). We adopted a similar approach to Study 2 in identifying the participants for Study 3. Samples were stratified by age, gender, and digital literacy to ensure comparability. Participants were randomly assigned to one of six chatbot versions:

1. Baseline: operationalised through default English phrasing, no cultural tailoring.
2. Enculturation: operationalised through localised greetings, idioms, and culturally familiar salutations.
3. Adaptive Interaction: operationalised through pacing, rhythm, and conversational cues adjusted to the user's style.
4. Explainability and Transparency: operationalised through culturally familiar metaphors, narrative-style explanations, and clear justifications.
5. Governance and Accountability: operationalised through explicit fairness cues, privacy assurances, and statements about data responsibility.
6. The full CRAIF-C implementation with all four domains deployed simultaneously.

Similar to Study 2, all interactions occurred in a secure apparel e-commerce shopping sandbox that mirrored production customer-journey workflows (e.g., product discovery and comparison, size/fit guidance, promotions and checkout, order status and returns).

A manipulation check (Oppenheimer, 2009) confirmed that more than 85 % of participants correctly identified their assigned chatbot's distinguishing feature, confirming manipulation fidelity. After the interaction with the respective chatbot, participants completed the CRAIF-C domain scales (five items per domain; see Appendix 3). Confirmatory factor analysis supported construct validity (CFI = 0.95, RMSEA = 0.042, AVE > 0.70), and internal consistency was excellent (Cronbach's $\alpha = .88$ –0.92). Multi-group invariance testing confirmed equivalence across countries.

A MANOVA revealed a significant main effect of chatbot condition on the dependent variable: usability, measured with the System

Usability Scale (Brooke, 1996). Wilks' Lambda = 0.64, $F(20, 2396) = 18.42$, $p < .001$, $\eta^2 = 0.23$. Univariate tests showed each CRAIF-C domain significantly outperformed the baseline on its respective measure with effect sizes varying across countries, reflecting cultural differences in which aspects of responsiveness were most valued:

- Enculturation: $\Delta M = +1.30$, 95 % CI [1.12, 1.48], $p < .001$, $\eta^2 = 0.27$; largest gains in India (+1.30, $p < .001$) and the Philippines (+1.10, $p < .001$).
- Adaptive Interaction: $\Delta M = +1.20$, 95 % CI [1.03, 1.37], $p < .001$, $\eta^2 = 0.25$; most valued in Vietnam (+1.20, $p < .001$) and the Philippines (+1.10, $p < .001$).
- Explainability and Transparency: $\Delta M = +1.40$, 95 % CI [1.23, 1.57], $p < .001$, $\eta^2 = 0.30$; strongest effect among Australian participants (M from 3.10 to 4.50, $p < .001$).
- Governance and Accountability: $\Delta M = +1.10$, 95 % CI [0.94, 1.26], $p < .001$, $\eta^2 = 0.24$; most influential in Vietnam (+1.10, $p < .001$).
- Full CRAIF-C implementation: $\Delta = +11.9$, 95 % CI [9.6, 14.2], $p < .001$, representing a 16.3 % improvement in usability from 72.8 in the baseline to 84.7.

Similar to Study 2, we assessed robustness with subgroup analyses by age (18–29, 30–44, 45+) and gender within each locale (Australia, India, Philippines, Vietnam). For each CRAIF-C domain vs Baseline contrast, we estimated Hedges' g with 95 % CIs by subgroup and ran follow-up ANCOVAs (HC3) including Condition $\times$ Age and Condition $\times$ Gender ($\alpha = .05$; Holm-adjusted). Effects were directionally consistent and statistically reliable across subgroups (CIs excluded 0; adjusted $p < .05$), with overall magnitudes in the moderate range ($g \approx 0.31$ –0.51). The strongest subgroup gain was in the Philippines 18–29 ($g = 0.51$, 95 % CI [0.37, 0.65]); the smallest was in Vietnam 45+ ($g = 0.31$, [0.18, 0.44]). Gender effects were near-identical within locales (men ≈ 0.34 –0.49; women ≈ 0.35 –0.50). Interaction tests were nonsignificant after Holm correction (Condition $\times$ Age: $p = .41$ –0.77; Condition $\times$ Gender: $p = .49$ –0.69), indicating comparable gains across age and gender within each locale.

Open-ended responses ($n = 298$) echoed the quantitative findings, with participants describing the chatbots as “familiar,” “clear in reasoning,” and “trustworthy.” Comments mapped directly to the framework's domains, reinforcing construct validity of the CRAIF-C domain scale and the real-world usability benefits of culturally responsive design.

In sum, the findings from Study 3 demonstrate that no single CRAIF-C domain is universally “more important” across the cultural contexts examined; rather, the value of each domain is shaped by the cultural expectations and communication norms of specific user groups. Nonetheless, Explainability and Transparency stands out as the domain with the largest single-condition effect size overall ($\eta^2 = 0.30$), producing the most substantial average improvement (+1.40) in user experience. This effect was most pronounced among Australian participants, a lower-context culture relative to the others in the study, where communication norms emphasise explicitness, clarity, and direct reasoning. In such environments, explanation-rich interactions reduce cognitive load and uncertainty, aligning with user expectations for transparent, step-by-step justification in decision-making systems. Clear, culturally familiar explanations not only improve comprehension but also act as signals of competence, fairness, and accountability, attributes that directly influence trust and adoption. Other domains showed strong but more context-specific impacts: Enculturation (+1.30, $\eta^2 = 0.27$) resonated most in India, where formal salutations and relational cues enhanced perceived authenticity. Adaptive Interaction (+1.20, $\eta^2 = 0.25$) was most valued in Vietnam and the Philippines, where conversational pacing and metaphorical reasoning shaped engagement. Finally, Governance and Accountability (+1.10, $\eta^2 = 0.24$) proved particularly influential in building trust in Vietnam. Notably, the highest overall usability scores (System Usability Scale - SUS) were achieved only under

full CRAIF-C integration, suggesting that while individual domains can be strategically prioritised for regional deployment, the framework's greatest strength lies in the synergistic effect of its combined components.

4.4. Study 4: naturalistic deployment of CRAIF-C

The first three studies were conducted in controlled or semi-controlled environments, enabling precise measurement but not fully reflecting the complexity and pressures of commercial operations. Study 4 extended this work by examining CRAIF-C in live, high-traffic deployments over a twelve-month period. The aim was to determine whether the framework's effects on trust, engagement, and business outcomes could be sustained in real-world conditions.

Two corporate clients of PolyglotAI participated in Study 4: a fintech company and a large-scale e-commerce platform, both operating in Indonesia. Prior to this intervention, each had run generic multilingual chatbots for over a year and had received consistent negative customer feedback about tone, cultural alignment, and trust. During the 12-month deployment, participants were real customers of the two Indonesian clients (fintech; large-scale apparel e-commerce). Recruitment used in-product intercepts (website/app banner at chat launch and post-chat survey cards) and email/SMS invitations to recent users who had consented to be contacted for research. Eligibility required age ≥ 18 , residency in Indonesia, fluency in Bahasa Indonesia (English optional), and at least one prior chatbot interaction in the last 6 months. We implemented device/geo checks and duplicate suppression (IP/device fingerprinting) and excluded suspected bot traffic and VPN/proxy use. Sampling was quota-balanced by device type (mobile/desktop), broad age band, and gender to reflect each client's user base. After an explicit just-in-time consent screen, participants were block-randomised by client and language preference; returning participants were re-contacted at pre-assigned intervals (T1/T2) using pseudonymous IDs to preserve privacy. Free-text was automatically de-identified and manually screened for sensitive information; no payment or account identifiers were collected.

During the baseline phase (T0), each chatbot remained in its standard configuration without cultural adaptations. Over the next twelve months, both systems were upgraded to full CRAIF-C implementations, integrating all four domains directly into conversational architecture and user flows. Enhancements included localised language models, culturally adapted greetings and idioms, adaptive pacing rules, culturally familiar explanatory metaphors, and accountability signals such as explicit consent checks and fairness prompts (see Appendix 4).

Data were collected at three points ($n = 200$; 94 from the fintech company and 106 from the e-commerce platform): baseline (T0), six months post-deployment (T1), and twelve months post-deployment (T2). Perceptual measures were drawn from the same validated, CRAIF-C domain-specific scales used in earlier studies. Cronbach's alpha values ranged from 0.86 to 0.92 across domains and time points, confirming strong internal reliability.

A repeated-measures MANOVA showed significant main effects of time across all domains, Wilks' Lambda = 0.54, $F(8, 1094) = 109.2$, $p < .001$, $\eta^2 = 0.28$. Gains from T0 to T1 were substantial ($p < .001$) and were either sustained or further improved by T2. Table 5 presents the domain-level means, standard deviations, 95 % confidence intervals, p-values, and effect sizes.

Table 5
Changes in CRAIF-C domain scores over time (T0, T1, T2).

Domain	T0 Mean (SD)	T1 Mean (SD)	T2 Mean (SD)	95 % CI for Change (T0–T2)	p-value	η^2
Enculturation	3.32 (0.58)	4.15 (0.54)	4.27 (0.51)	[0.82, 1.05]	<0.001	0.26
Adaptive Interaction	3.41 (0.61)	4.19 (0.55)	4.29 (0.53)	[0.76, 1.00]	<0.001	0.24
Explainability & Transparency	3.28 (0.63)	4.18 (0.57)	4.35 (0.50)	[0.88, 1.12]	<0.001	0.27
Governance & Accountability	3.36 (0.60)	4.13 (0.56)	4.25 (0.54)	[0.75, 0.99]	<0.001	0.23

We further tested whether improvements from T0 $\rightarrow$ T1 $\rightarrow$ T2 varied by domain (fintech and apparel e-commerce), age (18–29, 30–44, 45+), and gender using a repeated-measures MANCOVA (Time within-subjects; Domain/Age/Gender between-subjects; prior chatbot experience as covariate). The overall time effect remained robust (Wilks' $\Lambda = 0.54$, $F(8, 1094) = 109.2$, $p < .001$, partial $\eta^2 = 0.28$). Interaction tests showed no differential trajectories: Time $\times$ Domain, Time $\times$ Age, and Time $\times$ Gender were not significant after Holm adjustment (all $p_{adj} > 0.05$; partial η^2 in the negligible–small range). Within each subgroup, pairwise gains from T0 $\rightarrow$ T1 and T0 $\rightarrow$ T2 were statistically reliable across all CRAIF-C scales (two-sided tests, Holm-adjusted $p < .01$), with Hedges' g for T0 $\rightarrow$ T2 in the small-moderate range and 95 % CIs excluding 0.

Behavioural metrics from chatbot logs also supported the perceptual findings. Average session duration rose from 2.3 min at T0 to 3.5 min at T1 and 3.7 min at T2, a 61 % increase, $F(2, 12480) = 34.7$, $p < .001$. Conversation depth increased from 6.1 to 9.4 to 9.8 turns per session (+60.6 %, $p < .001$). Repeat user rates grew from 18.6 % at T0 to 27.3 % at T1 and 29.4 % at T2 ($\chi^2 = 45.2$, $p < .001$). Click-through rates on recommended items rose from 14.2 % to 24.1 %–24.9 % (+10.7 %, $p < .001$), while purchase conversions grew from 6.5 % to 10.9 %–11.3 %, a 73.8 % relative increase ($p < .001$).

Sentiment analysis of open-ended feedback further revealed higher cultural appreciation over time: positive sentiment increased from 48 % at T0 to 71 % at T1 and 76 % at T2. Users praised familiar language, natural pacing, clear explanations, and respectful governance cues, aligning with the four CRAIF-C domains. Table 6 provides a representative sample of the collected feedback.

These qualitative findings complement the quantitative data, reinforcing the conclusion that cultural responsiveness influences not only user satisfaction but also trust and transactional behaviour.

Overall, the findings from Study 4, show that CRAIF-C can deliver sustained perceptual and behavioural benefits in live commercial contexts. These improvements were not superficial but reflected deeper cultural alignment that enhanced both user trust and business performance over time.

Table 6
Representative user feedback illustrating perceived benefits of CRAIF-C.

CRAIF-C Domain	Representative User Quotes
Enculturation	“It used phrases I grew up hearing, like it knew how we think.” “Even small things like the way it greeted me felt more natural. It didn't feel robotic or foreign.”
Adaptive Interaction	“The pace of the conversation was just right. It didn't rush me or push products too fast.” “It adjusted based on how I was replying. It didn't repeat itself like other bots.”
Explainability & Transparency	“I liked that it told me why it was recommending something. It wasn't just a random list. It made sense.” “When it explained things using examples I understood, I felt more confident clicking through.”
Governance & Accountability	“It felt respectful and well-designed, like someone thought about how I would actually use it.” “I appreciated that it asked for my preferences instead of assuming them. It felt like I had control.”

5. Discussion

Across Studies 1–4, the pattern of results aligns with CRAIF-C's theorised mechanisms and strengthens attribution to its specific components. Study 1 establishes the foundation showing that Enculturation, Adaptive Interaction, Explainability & Transparency, and Governance & Accountability map onto distinct user outcomes. Study 2 (baseline vs four CRAIF-C domains; MANCOVA) demonstrates selective improvements consistent with those mechanisms: Enculturation yields the largest gains on Pragmatics-Fit and meaningful lifts in Trust/Appropriateness; Adaptive Interaction improves SUS and perceived appropriateness (smoother turn-taking and pacing); Explainability & Transparency raises explainability ratings without degrading factuality (one-sided 95 % CIs satisfy $\Delta \leq 0.02$); and Governance & Accountability increases perceived accountability while reducing cultural-harm flags and preserving fairness across locale, age, and gender. Study 3 (single-component ablations) corroborates selective causality: turning a component off attenuates its targeted outcome with minimal spillover to non-targets, mirroring the component-vs-baseline lifts in Study 2. Study 4 (12-month deployment in fintech and apparel e-commerce) shows these gains persist and generalise in real-world use, with stable or improving trajectories over time and safety maintained (factuality non-inferiority and no subgroup fairness degradation). Together, the four studies move from foundational aspects (Study 1) to experimental attribution (Studies 2–3) to field robustness (Study 4), linking observed improvements directly to the specific components of CRAIF-C.

5.1. Theoretical contributions

This research advances work in human-computer interaction (HCI) and AI ethics by demonstrating how the Culturally Responsive Artificial Intelligence (Chatbot) Framework (CRAIF-C) can be structurally embedded into AI systems as a primary architectural principle, rather than as an optional surface layer. Across four interlinked studies, CRAIF-C's domains (Enculturation, Adaptive Interaction, Explainability and Transparency, and Governance and Accountability) were implemented, measured, and refined in both controlled and live deployment settings.

The first contribution of this work is the repositioning of enculturation from a peripheral localisation task to a central architectural principle. Evidence from Studies 2, 3, and 4 shows that AI systems using culturally congruent communication styles, narrative structures, and tonal patterns consistently produced higher trust and satisfaction ratings (improvements of +1.1 to +1.4 points, all $p < .001$). This operationalises earlier HCI theory on the role of cultural norms in adoption (Alkhamissi et al., 2024; Hall, 1976; Marcus & Gould, 2000; Yan et al., 2024) by embedding them directly into the model's interaction logic.

The second contribution is reframing adaptive interaction as a dynamic, culturally modulated capability. CRAIF-C moves beyond static or user-initiated adjustments toward culturally modulated conversational pacing, turn-taking, and prompting. Studies 2 and 3 demonstrated significant gains in conversational naturalness ($\eta^2 = 0.12$ – 0.25), and Study 4 showed these gains persisted for at least twelve months. This supports the positioning of adaptivity as a core alignment mechanism in multi-cultural AI deployments.

The third contribution reframes explainability as a culturally contingent construct. Study 3 found that high-context cultures favoured metaphorical or story-based explanations, while low-context cultures preferred step-by-step analytical formats. These differences were stable across controlled and live environments (Studies 3 and 4), challenging the assumption in much of the explainable AI literature that interpretability is universal (Peters & Carman, 2024).

The fourth contribution is operationalising governance and accountability through pluralistic, culturally literate structures. CRAIF-C translates abstract ethical commitments into concrete operational processes. In Study 1, a start-up closed governance gaps through co-design and fairness metrics. Study 4 extended this to commercial

contexts, where governance cues such as explicit consent checks improved trust and increased repeat usage by over 10 percentage points. This aligns with calls to move AI ethics from high-level principle statements to culturally legitimate governance practices (Fjeld et al., 2020; Jobin et al., 2019).

Taken together, these contributions position CRAIF-C as a framework aligned to risk-based regulatory guardrails which many jurisdictions are embedding into AI deployments. The 2024 EU Artificial Intelligence Act, for example, introduces tiered risk duties and explicit transparency obligations for AI systems such as conversational agents and chatbots (European Parliament & Council of the European Union, 2024; EUR-Lex, 2024). While the Act does not single out culture, its risk-management logic could be extended to context-specific failure modes, including those arising from cultural misalignment, making culturally aware design and evaluation a prudent component of compliance and accountability in cross-border deployments.

Concurrently, engineering guidance has matured: NIST's AI Risk Management Framework (AI RMF 1.0) now has a Generative AI Profile (NIST-AI-600-1) that operationalises governance-map-measure-manage practices for genAI. These practices align with concrete hooks for identifying culture-linked risks, building locale-specific evaluations, and instituting mitigation and review gates (NIST, 2024). Empirically, past work have shown that multilingual capability does not guarantee multicultural alignment with recent evaluations show large language models (LLMs) tend to mirror value profiles closest to English-speaking and Protestant European contexts by default, with cultural prompting improving, but not eliminating, misalignment across many locales (Yan et al., 2024). Newer work also systematically compares model outputs with World Values Survey distributions, again finding gaps between language proficiency and cultural fit (Ryström et al., 2025; Yan et al., 2024). In parallel, explainable AI (XAI) research has been shown to over-sample Western participants and to privilege analytic, feature-attribution styles of explanation, potentially clashing with narrative or relational preferences in high-context cultures, evidence that explanation form itself is culturally situated (Peters & Carman, 2024). Fairness discourse has likewise shifted toward socio-technical remedies: recent surveys and scoping reviews argue that statistical constraints must be complemented by procedural and relational safeguards (e.g., participatory auditing, data provenance, and culturally situated evaluation) so that fairness is defined in context rather than presumed universal (Luo et al., 2024).

These developments have three concrete implications for our framework. First, localisation must extend beyond translation. Models can be fluent while mishandling pragmatics central to high-context communication (politeness, indirectness, taboo lexicons, honorifics), motivating culturally tagged corpora, locale-specific test sets, and human-in-the-loop workflows for high-risk intents; our Enculturation and Localisation components should be positioned as controls, not enhancements (Ryström et al., 2025). Second, evaluation must measure culture directly. We suggest value-congruence probes (topic-specific agreement distributions compared to local benchmarks), pragmatics tests (indirectness/politeness), and harm-benefit asymmetry checks for culturally salient errors, aligning our evaluation gates with the NIST RMF "MEASURE/MANAGE" functions (NIST, 2024). Third, explanations should be culturally configurable. Building on the XAI evidence base, we advocate for explanation-modality choices (narrative exemplars, value-anchored rationales, or analytic summaries) and recording the rationale in cultural model cards to support both usability and auditability (Peters & Carman, 2024). To aid adoption, we further propose that artefacts (cultural model cards, locale risk registers, explanation-modality rationales, and escalation runbooks) are mapped to NIST RMF functions and, where applicable, to EU AI Act transparency/risk provisions, making cultural responsiveness auditable for regulators and reproducible for practitioners (European Parliament & Council of the European Union, 2024; EUR-Lex, 2024; NIST, 2024). In short, our findings add to the AI/HCI literature, reinforcing that

language proficiency is insufficient for global readiness. Rather, cultural evaluation and governance must be first-class design requirements, a stance our framework now operationalises through lifecycle controls, measurable outcomes, and regulatory mapping.

5.2. Practical implications

The findings from the four studies offer actionable insights for AI leaders, product managers, and organisations aiming to embed cultural responsiveness into intelligent systems. The evidence shows that even small, targeted cultural adaptations such as aligning tone, greetings, idiomatic expressions, and symbolic cues can generate significant improvements in user trust and perceived relevance. In Studies 2 and 3, simple adjustments to opening phrases that reflected relational norms produced measurable gains in both perceived fit and trust, while Study 4 demonstrated that these benefits persisted over time, sustaining engagement and repeat usage. Importantly, such adaptations can be implemented without the need for complete system redevelopment, which makes them feasible even in resource-constrained environments.

The research also confirms that culturally adaptive dialogue design consistently improves user satisfaction and conversation depth. Adjustments to pacing, turn-taking, and conversational cues in Studies 2 and 3 led to immediate improvements, and Study 4 verified that these enhancements maintained their impact in real-world deployments. This suggests that adaptivity should be treated as a core design principle, one that can be introduced incrementally according to market and cultural priorities.

Another key finding is that explanation formats should match cultural communication styles in order to build trust effectively. In Study 3, narrative and metaphorical explanations performed better with high-context cultures such as India, Vietnam, and the Philippines, while analytical formats were more effective in low-context cultures. Study 4 confirmed that these culturally aligned explanation strategies maintained comprehension and user confidence in live systems. This reinforces that interpretability must be tailored to cultural expectations rather than treated as uniform across audiences.

Finally, the results show the strategic value of embedding cultural responsiveness into governance structures. Study 1 illustrated how small start-ups could achieve this through cultural champions, partnerships, and modular architecture. Study 4 showed that corporate clients improved both ethical credibility and business performance by integrating cultural governance cues into their AI systems. Taken together, the evidence demonstrates that CRAIF-C is a scalable and modular framework that can be adapted to diverse contexts, delivering measurable benefits when cultural responsiveness is built into the design and governance of AI systems rather than added superficially.

6. Limitations and future directions

While this research offers robust empirical evidence and substantive theoretical contributions, several limitations must be acknowledged. These, in turn, open up valuable avenues for future work.

The first limitation relates to the scope of Study 1, which was confined to a single AI development team in a small start-up environment. This context provided the advantage of deep ethnographic immersion and fine-grained observation of CRAIF-C's operationalisation. However, the size, disciplinary composition, and regional location of the team inevitably constrain the generalisability of the findings. Larger, more heterogeneous environments, such as multinational corporations, geographically distributed open-source communities, or public sector agencies, are likely to encounter different adoption barriers, governance pressures, and pathways of cultural integration.

Second, our evaluation focused on text-based chatbot systems in e-commerce; hence, the hyphenated "C" in CRAIF-C. While this setting is controlled and commercially relevant, generalisability to other AI modalities remains to be established. Voice assistants, embodied/robotic

agents, and vision-centric applications introduce different cultural cues (e.g., prosody, gesture, gaze, visual symbolism) and governance needs. We envisage modality-specific extensions: e.g., CRAIF-V (voice), CRAIF-MM (multimodal/vision), CRAIF-E (embodied), and sectoral variants for healthcare, education, and public services. Testing these variants in domain-appropriate tasks will be essential to demonstrate cross-modal and cross-sector robustness.

A third limitation concerns the temporal scope of Study 4. Although the longitudinal design provided a richer view than short-term trials, the follow-up period was limited to twelve months. Gains in trust, engagement, and conversion were sustained across this timeframe, but it is unclear whether such effects would persist, plateau, or diminish over multiple years. Longer-term studies could also investigate how shifts in cultural norms, particularly in fast-globalising or digitally saturated contexts, alter the interpretation and salience of CRAIF-C's domains.

The fourth limitation involves measurement. While the scales we adopted to measure the CRAIF-C domains demonstrated strong reliability and construct validity in the settings studied, they cannot be assumed to be universally stable. For example, cultural responsiveness is not only about differences between national or linguistic groups. It also varies within them. Intra-cultural differences, such as those linked to regional identities, socio-economic backgrounds, generational cohorts, or professional subcultures, can significantly influence how users perceive tone, pacing, symbolism, and governance cues. Instruments validated at a broad cultural level may therefore overlook meaningful within-group variation, leading to partial or misleading conclusions. As CRAIF-C is extended into new cultural, linguistic, and technological contexts, it will require iterative refinement to capture both cross-cultural and intra-cultural nuances. Mixed-method strategies, such as co-design workshops, participatory evaluation, and the use of cultural probes, can help surface emergent constructs and context-specific indicators that are not easily captured by fixed-format scales, ensuring the framework remains sensitive to the full spectrum of cultural diversity it aims to serve.

Fifth, while our design (Studies 2–4) links improvements to CRAIF-C components, we cannot rule out alternative explanations (e.g., novelty/Hawthorne effects, concurrent product changes, seasonality, or unmeasured user characteristics) that may have contributed to the observed gains. Ablations strengthen attribution but do not fully eliminate residual confounding, especially in the 12-month field setting. Future research should use component-level randomised field experiments (e.g., stepped-wedge or crossover A/B tests), pre-registered mediation analyses that trace component → mechanism → outcome, and instrumentation that logs when a CRAIF-C control is invoked. Multi-site replications across additional locales and sectors, with negative-control outcomes and stricter blinding of raters, would further test causality and generalisability while monitoring for drift and secular trends.

Sixth, our findings should be read in light of several methodological limitations, which, in turn, present opportunities for future research. For instance, the recruitment via online panels (Study 2) and in-product intercepts (Study 4) can introduce self-selection, coverage (device/geo), and non-response bias. Our use of blocked randomisation and re-weighting reduces, but does not eliminate, these risks. Similarly, the cultural and domain scope is narrow: Study 2 focuses on apparel e-commerce in Indonesia, Vietnam, and Thailand, and Study 4 evaluates two Indonesian clients (fintech and e-commerce). These choices improve ecological relevance but limit transportability to other cultures, languages, and sectors.

Finally, scaling culturally responsive AI requires alignment with governance frameworks and regulatory standards that ensure consistent application across contexts. This alignment is especially critical in high-stakes domains such as healthcare, finance, and public administration, where regulatory oversight is tightening, and failures in cultural responsiveness can undermine trust, incur legal liabilities, and cause tangible harm. Future studies could explore pathways for embedding

CRAIF-C principles into certification processes, sector-specific guidelines, and international AI governance frameworks, testing how such integration can transform cultural responsiveness from an aspirational value into a measurable and enforceable standard.

7. Conclusion

This paper introduced and validated the Culturally Responsive Artificial Intelligence (Chatbot) Framework (CRAIF-C) as both a theoretical model and a practical approach for embedding cultural responsiveness into AI systems. Drawing on four interconnected studies, it has shown that cultural responsiveness can be operationalised as a measurable driver of trust, engagement, usability, and business value in human-AI interaction. By embedding cultural justice alongside technical fairness, CRAIF-C addresses a critical gap in AI development by offering researchers, developers, and policymakers a practical roadmap for designing systems that resonate with local norms, respond empathetically to users, and incorporate diverse epistemologies.

As AI continues to cross borders and enter culturally diverse markets, CRAIF-C provides timely guidance for making cultural responsiveness a first-order design consideration. The studies presented here demonstrate how the framework can be translated from theory to practice, from controlled research settings to sustained real-world deployment. In doing so, CRAIF-C offers a pathway for creating AI that is not only technically capable but also socially attuned and ethically robust in a global digital ecosystem. Such a pathway aligns with global policy priorities such as UNESCO's Recommendation on the Ethics of Artificial Intelligence (UNESCO, 2021), which calls for culturally inclusive and human-centred AI governance.

Taken together, the findings of this study speak to several long-running themes in AI and HCI literature. First, classic HCI work on "quality in use" positions usability within a broader frame that includes effectiveness, satisfaction, and freedom from risk (International Organization for Standardization, 2018; International Organization for Standardization; International Electrotechnical Commission, 2023). Our findings add a culture-first layer to that canon: by treating culture as a first-class requirement and instrumenting it (enculturation assets, pragmatics probes, explanation style), we show consistent gains in system usability, trust/appropriateness, and pragmatics-fit while holding safety constant (factuality non-inferiority and no subgroup fairness degradation). This moves beyond generic interface tuning toward auditable, locale-aware controls embedded in the development lifecycle.

A second theme is alignment and safety in language models. Prior work shows that preference learning can reduce harmful or off-target behavior and improve perceived helpfulness (RLHF/Instruction-tuning: Ouyang et al., 2022; policy-guided harmlessness: Bai et al., 2022), and that diverse, scenario-rich evaluation is needed to surface failure modes (HELM: Liang et al., 2022; toxicity benchmarks: Gehman et al., 2020). We complement these algorithmic approaches with socio-cultural alignment: component-level results (Study 2) and ablations (Study 3) indicate that enculturation, adaptive interaction, explanation, and governance each produce selective,

mechanism-consistent improvements without sacrificing factuality. The 12-month field evidence (Study 4) further speaks to deployment robustness addressing the literature's calls for longitudinal validation of aligned systems (Kjærup et al., 2021; Long et al., 2025).

A third theme concerns transparency, fairness, and documentation. Model Cards argued for disaggregated reporting and context on intended use (Mitchell et al., 2019); work on demographic performance disparities underscored the necessity of subgroup analysis (Buolamwini & Gebru, 2018). Our design operationalises these recommendations with pre-registered subgroups by locale, age, and gender; governance and accountability cues inside the dialogue; and decision logs that map to risk-management guidance (e.g., NIST AI RMF 1.0). Empirically, fairness does not degrade when culture-first adaptations are introduced, an important complement to prior findings that safety interventions can sometimes trade off with utility.

Finally, cross-cultural HCI has long emphasised localisation, translation quality, and cultural heuristics, but often without end-to-end pipelines that tie design moves to measurable outcomes (Aykin, 2005; Del Galdo and Nielsen, 1996; Marcus & Gould, 2000; Sun, 2012). By coupling standard localisation methods (back-translation, cognitive debriefing) with preference learning and policy gating, we provide a reproducible, gate-based process that links cultural mechanisms (tone, pacing, explanation style, governance signals) to validated scales and real-world metrics. In doing so, the studies help bridge human-centred design, evaluation science, and modern alignment methods. They show that culturally responsive conversational AI can deliver user-relevant improvements that are statistically reliable, operationally meaningful, and governed in line with contemporary standards for trustworthy AI.

CRedit authorship contribution statement

Vik Naidoo: Writing – original draft, Project administration, Methodology, Investigation, Data curation, Conceptualization. **Karman Kaur Chadha:** Writing – review & editing, Validation, Methodology, Formal analysis.

Declaration of competing interest

The authors disclose that the paper has undergone an AI-assisted edit check through Grammarly version 9.8, while preserving the original intent and content.

The authors have no relevant financial or non-financial interests to disclose.

The authors have no competing interests to declare relevant to this article's content.

The authors confirm that this study used only fully de-identified data, with no reasonable risk of re-identification. In accordance with Clause 5.1.17 of the Australian Government's National Statement on Ethical Conduct in Human Research (National Institute of Standards and Technology, 2023), such use of non-identifiable data does not require review by a Human Research Ethics Committee.

Appendix A Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.chbah.2025.100224>.

Appendix 1. CRAIF-C Implementation Blueprint

Domain	Goal (What to achieve)	Key Artefacts	Implementation Steps	Metrics/KPIs	Risks & Controls	Governance/Docs
Enculturation	Encode market-specific norms so the bot “speaks the culture”	Cultural ontology; tagged corpora; taboo/consent lexicons; prompt templates; coverage dashboard	(1) Define cultural variables (greetings, face-work, formality); (2) Collect/curate texts; (3) Back-translate + local expert annotation; (4) Tag with market/domain/register; (5) QA & IAA checks; (6) Version and release assets	Coverage (% intents with cultural tags); IAA (κ/α); taboo leak rate; localisation turnaround	Bias in sources → diversify; drift → quarterly reviews; low agreement → adjudication clinics	Datasheets for datasets; Ontology changelog; CIA pre-approval notes
Adaptive Interaction	Adjust style and pacing to user/culture while preserving task success	Reward functions with cultural terms; policy checkpoints; style controllers	(1) Bandit warm-start for style; (2) Constrained RLHF with cultural rewards; (3) Rate-limited online adaptation; (4) Human-in-the-loop escalations	Task success; Tone/Pacing Fit; Face-threat incidents; Avg turns; Escalation rate/latency	Value drift → cap updates; over-personalisation → guardrails; safety conflicts → safety overrides	Policy card (objectives, weights); Audit logs; Incident escalation SOP
Explainability & Transparency	Provide explanations that resonate culturally	Explanation templates (analytic & narrative); localised exemplars; source stubs	(1) Offer user-selectable mode; (2) Localise examples/idioms; (3) Length controls; (4) Log explanation usage	ERI (Explanation Resonance Index); Helpfulness; Trust delta pre/post explanation	Over-disclosure → redact sources; cognitive load → length caps	XAI catalogue; Approved exemplar library; Panel audit records
Governance & Accountability	Make accountability plural and auditable	CIA template; Extended model cards; RACI; Incident taxonomy; Rollback criteria	(1) Constitute co-design/panel; (2) Run CIA; (3) Define RACI & SLAs; (4) Log/audit; (5) Drill rollback; (6) Quarterly review	Incident frequency/severity; Time-to-rollback; Panel quorum; Action closure rate	Ownership gaps → RACI; slow rollback → runbooks; compliance gaps → residency/legal checks	Model card (cultural coverage, known gaps); CIA report; Quarterly governance minutes

Note: QA = Quality Assurance; IAA = Inter-Annotator Agreement (e.g., Cohen’s κ); RLHF = Reinforcement Learning from Human Feedback; ERI = Explanation Resonance Index; CIA = Cultural Impact Assessment; RACI = Responsible–Accountable–Consulted–Informed; SOP = Standard Operating Procedure; SLA = Service Level Agreement; KPI = Key Performance Indicator; XAI = Explainable AI.

Appendix 2. Original 12 Themes

Axial Theme	Explanation
1. Shifting Design Consciousness	Developers began to critically reflect on how dominant (often Western) norms were embedded in their interface and model design assumptions.
2. Localisation vs. Scalability	Tension between tailoring AI outputs for cultural nuance and the need to maintain code modularity or scalability across markets.
3. Cultural Empathy in Code	Recognition that cultural awareness is not just UX-layered but must inform architectural and interaction design logic.
4. Governance Through Partnerships	The need to bring in external cultural experts, NGOs, and users to collaboratively define fairness and appropriateness.
5. Modularisation for Sustainability	Creation of culturally adaptable templates and prompts to streamline future localisations without rewriting code.
6. Affective Transformation	Emotional shifts experienced by the team: greater empathy, ethical awareness, and personal responsibility for inclusivity.
7. Reframing Error Diagnosis	The team began attributing chatbot “failure” not just to bugs but to cultural disconnects, reshaping how QA was conducted.
8. Participatory Friction	Challenges encountered when integrating external reviewers: time delays, conflicting feedback, and boundary negotiation.
9. Re-anchoring Explanation Logic	Teams rethought how chatbot explanations should be structured: moving from logic-heavy outputs to narrative or metaphor.
10. Cultural Debugging	A new process where test failures were traced back to cultural misalignment in prompts, tone, or context, not just function.
11. Design Ownership Diffusion	A shift from cultural responsibility being the UX designer’s task to a shared ethos across roles (e.g., ML, Dev, Ethics).
12. Code-Level Resistance and Buy-in	Initial reluctance among engineers to prioritise cultural input, followed by gradual acceptance as impact became clear.

Appendix 3. CRAIF-C Domain Scale Items

Domain	Item Code	Scale Item	Source Adaptation
Enculturation	CE1	The chatbot’s language feels natural for my culture.	Reinecke and Bernstein (2013)
	CE2	The tone of the chatbot matches the way people speak in my community.	Reinecke and Bernstein (2013)
	CE3	The expressions and idioms used are familiar to me.	Reinecke and Bernstein (2013)
	CE4	The chatbot greets me in a culturally appropriate way.	Reinecke and Bernstein (2013)
	CE5	The interaction reflects an understanding of my cultural norms.	Reinecke and Bernstein (2013)
Adaptive Interaction	AI1	The chatbot adjusts its pace to match my communication style.	Jain et al. (2018)
	AI2	The chatbot changes its responses depending on how I interact.	Jain et al. (2018)
	AI3	The conversation flows naturally without feeling rushed or too slow.	Jain et al. (2018)
	AI4	The chatbot adapts to my preferred level of formality.	Jain et al. (2018)
	AI5	The system feels responsive to my conversational cues.	Jain et al. (2018)
Explainability & Transparency	ET1	The chatbot clearly explains the reasons behind its recommendations.	Kizilcec (2016); Hoffman et al. (2018)
	ET2	The explanations are easy for me to understand.	Kizilcec (2016)
	ET3	The chatbot uses examples that make sense to me.	Hoffman et al. (2018)
	ET4	The explanations fit with my own way of thinking.	Kizilcec (2016)
	ET5	I feel more confident because I understand why it made its recommendations.	Hoffman et al. (2018)

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(continued)

Domain	Item Code	Scale Item	Source Adaptation
Governance & Accountability	GA1	The chatbot communicates how it uses my data.	Jian et al. (2000); Glikson and Woolley (2020)
	GA2	The chatbot assures me that my data will be handled fairly.	Jian et al. (2000)
	GA3	The chatbot seems honest in its interactions.	McKnight et al. (2011); Jian et al. (2000)
	GA4	I believe the chatbot is designed to avoid bias.	Glikson and Woolley (2020)
	GA5	The chatbot makes me feel it is accountable for its actions.	Glikson and Woolley (2020)
User Engagement Scale	UE1	I lost myself in this experience.	O'Brien et al. (2018)
	UE2	The time I spent using Application X just slipped away.	O'Brien et al. (2018)
	UE3	I was absorbed in this experience	O'Brien et al. (2018)
	UE4	I felt frustrated while using this Application X (R)	O'Brien et al. (2018)
	UE5	I found this Application X confusing to use (R)	O'Brien et al. (2018)
	UE6	Using this Application X was taxing (R)	O'Brien et al. (2018)
	UE7	This Application X was attractive.	O'Brien et al. (2018)
	UE8	This Application X was aesthetically appealing	O'Brien et al. (2018)
	UE9	This Application X appealed to my senses	O'Brien et al. (2018)
	UE10	Using Application X was worthwhile	O'Brien et al. (2018)
	UE11	My experience was rewarding	O'Brien et al. (2018)
	UE 12	I felt interested in this experience	O'Brien et al. (2018)
System Usability Scale (SUS)	SU1	I think that I would like to use this system frequently	Brooke (1996)
	SU2	I found the system unnecessarily complex	Brooke (1996)
	SU3	I thought the system was easy to use	Brooke (1996)
	SU4	I think that I would need the support of a technical person to be able to use this system	Brooke (1996)
	SU5	I found the various functions in this system were well integrated	Brooke (1996)
	SU6	I thought there was too much inconsistency in this system	Brooke (1996)
	SU7	I would imagine that most people would learn to use this system very quickly	Brooke (1996)
	SU8	I found the system very cumbersome to use	Brooke (1996)
	SU9	I felt very confident using the system	Brooke (1996)
	SU10	I needed to learn a lot of things before I could get going with this system	Brooke (1996)

Appendix 4. Representative Shopping-Agent Dialogues (Six Markets)

Market	Scenario	Baseline (generic agent)	CRAIF-C response (encultured agent)	Markers	Why this (brief)
Indonesia (ID)	A. Product discovery (modest workwear)	"Here are blouses under \$30."	"Good afternoon, Ma'am. Modest office picks (long sleeves, opaque, relaxed fit): Linaria LS, Ayu Crepe Tunic, Kirana Cotton Top."	[E][A]	Filters: sleeve $\geq$ full, opacity high; price 200–350k IDR
Indonesia (ID)	B. Returns & courier	"Print a label and ship it."	"Book GoSend pickup or JNE drop-off, whichever's easier. Label in app; refund in 3–5 days."	[E][A] [X-N]	Local couriers reduce friction; practical guidance for Jakarta traffic.
Vietnam (VN)	A. Sandal size/fit	"38 fits most."	"VN 38 = 24.0–24.5 cm. Try Lotus Strap (adjustable). Free exchange 7 days."	[E][A] [X-A]	Uses local size mapping and comfort margin.
Vietnam (VN)	B. Deals & payment	"10 % off sitewide."	"Today: 12 % off with MoMo/ZaloPay; COD available outside promo."	[E][A] [X-N]	Aligns with popular e-wallet usage patterns.
Thailand (TH)	A. Temple-appropriate dresses	"Trending summer dresses."	"For temple visits: covered shoulders, $\geq$ knee length. Picks: Sawasdee Midi, Jasmine Wrap, Chao Phraya A-line.""	[E][A] [X-A]	Cultural coverage + breathable fabrics for heat/humidity.
Thailand (TH)	B. Delivery timing	"3–5 days."	"Bangkok: next-day by Kerry; Upcountry: 2–4 business days; lockers if you're out."	[E][A] [X-N]	City vs upcountry logistics expectations.
India (IN)	A. School backpack	"20L, many pockets."	"For school: 25 L, reinforced straps, rain cover. Picks: Deccan 25L, Nilgiri Pro. COD/UPI ok."	[E][A] [X-A]	Durability specs + local payment options.
India (IN)	B. Festive wear	"Here are kurtas."	"Cotton-silk with light embroidery; consider sizing up; sleeves tailored in 24–48 h"	[E][A] [X-N]	Festive comfort + quick alterations norm.
Philippines (PH)	A. Rain-ready sneakers	"Here are sneakers."	"For rainy season: water-resistant uppers, grippy soles. Picks: Tropical Shield, Baybay Runner."	[E][A] [X-A]	Slip resistance $\geq$ 0.6; quick-dry lining
Philippines (PH)	B. Payment & pickup	"Card or cash."	"GCash/Maya, COD, and store pickup (Ayala/SM). E-wallet promos auto-apply."	[E][A] [X-N]	Matches mall-pickup culture and e-wallet promos.
Australia (AU)	A. Road running shoes	"Top sellers."	"Road: Harbour 9 (neutral) vs Outback Pro (stability)."	[A][X-A]	Clear spec compare: drop, rebound; 30-day road-test return.
Australia (AU)	B. Sustainable basics	"Organic tees here."	"GOTS-certified cotton, 180 gsm, relaxed fit. Ship 2–4 biz days (metro), 3–6 regional."	[E][A] [X-N]	Sustainability cues + AU shipping norms.

Note 1: [E] Enculturation/[A] Adaptive pacing/[X-A] Analytic/[X-N] Narrative. Note 2: For localisation markers, Brand and service names (e.g., GoSend, JNE in Indonesia; MoMo/ZaloPay in Vietnam; GCash/Maya in the Philippines), as well as fulfilment options like COD (cash on delivery), are used as contextualised examples to reflect common, market-specific practices.

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